DATA ANALYSIS PROJECT

Traffic Safety and Accidents

OBSERVATORY

PRESENTED BY

Mohamed Sayed Said

- Python
- Excel Dashboards

Ahmed Fathy Kamal

- Power BI (DAX, Data Modeling, Dashboards)
- pptx

Basmala Ahmed Fouad

- Tableau (Data Modeling, Dashboards)
- Excel Pivot Tables

Abdullah Hassan Fathy

- Excel (Statistics, Data Cleaning, Modeling)
- SQL

Jana Ahmed Ramzy

- Documentation
- SQL
- Excel Pivot Tables

UNDER THE SUPERVISION OF

Dr. Amal Mahmoud

Presentation Agenda

01 Project Overview and Why We Selected It

02 4W's and How's and Data Exploration

03 Project Summary

Data Cleaning, Outlier Treatment, Feature Engineering, Descriptive Statistics, Data Modeling, Insights Derived, Dashboards

04 Excel

05 SQL

06 Python

07 Power BI

08 Tableau

09 Final Insights

01- Project Overview and Why We Selected It

Database Information

TOTAL COLLISIONS	TIME PERIOD	TOTAL CASUALTIES	VEHICLES INVOLVED	SEVERITY LEVELS	COLUMNS
503,475	2020–2024	640,522	920,692	3	99
accident records	5 years	casualty records	vehicle records	Fatal / Serious / Slight	attributes / 3 datasets

Problem Statement

Road traffic accidents remain a major public safety concern, causing significant human and economic losses every year. By analyzing collision, casualty, and vehicle datasets, this project aims to uncover accident patterns, identify the key factors influencing accident severity, and provide data-driven insights that support safer roads and more effective traffic management.

Problem Statement

Road traffic accidents remain a major public safety challenge, leading to thousands of fatalities, injuries, and substantial economic losses every year. By analyzing collision, casualty, and vehicle data, this project seeks to understand accident patterns, identify the factors influencing accident severity, and support informed decisions for improving road safety.

PROJECT OBJECTIVES

1. Identify the key factors contributing to road traffic accidents.
2. Analyze accident severity under different road and weather conditions.
3. Discover accident trends and high-risk locations.
4. Evaluate the impact of vehicle types on accident outcomes.
5. Generate actionable insights to improve road safety.

Why We Selected This Project

This project was selected because it provides a realistic and comprehensive dataset that reflects real-world transportation challenges while demonstrating practical data analytics techniques.

Integrates multiple related datasets (Collisions, Casualties, and Vehicles) for comprehensive analysis.

Demonstrates the complete data analytics workflow — data cleaning, modeling, analysis, and visualization.

Explores real-world accident data to uncover meaningful patterns and trends.

Transforms raw data into actionable insights that support evidence-based decision-making.

What Happened? – Road Safety Overview

Road Safety Authorities Observed:

- Road traffic collisions are concentrated in specific locations and road environments rather than being evenly distributed.
- Casualties range from slight injuries to fatal crashes, with injury severity varying according to road conditions and traffic characteristics.
- Urban roads experience the highest number of reported collisions due to higher traffic density and vehicle movement.

Key Findings

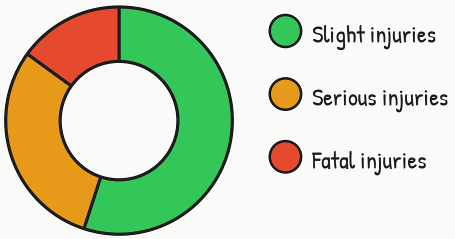
- Passenger cars are involved in the majority of reported collisions.
- Slight injuries account for the largest share of casualties, while fatal and serious injuries remain concentrated in high-risk conditions.
- A relatively small number of roads and locations generate a disproportionately high number of collisions.

Road Safety Insight: The collision data reveals clear spatial and operational patterns, indicating that targeted safety improvements can significantly reduce both accident frequency and casualty severity.

12,480
Total Collisions

18,650
Total Casualties

Casualties by Severity



Illustrative figures

Collision Concentration

Most road collisions occur repeatedly in urban areas and heavily trafficked roads.

Casualty Severity

Serious and fatal injuries are mainly associated with high-speed roads and vulnerable road users.

Why Did It Happen? – Root Cause Analysis

Key Contributing Factors: Several environmental, road, and driver-related factors contribute to road collision occurrence and severity.

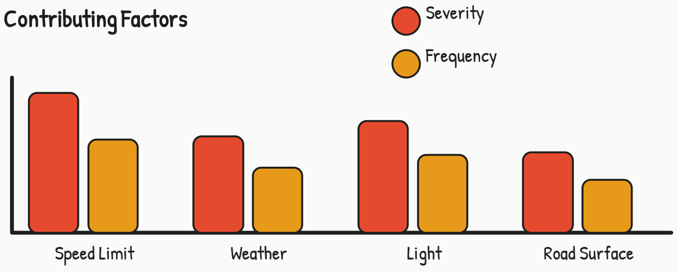
High-Risk Factors

- High-speed roads increase collision severity.
- Reduced visibility during nighttime contributes to more serious crashes.
- Wet road surfaces and adverse weather conditions reduce vehicle control.
- Junctions and complex vehicle manoeuvres increase collision likelihood.
- Motorcyclists and pedestrians remain the most vulnerable road users.

Protective Factors

- Lower speed limits reduce crash severity.
- Good lighting improves driver visibility.
- Well-maintained roads reduce collision risk.
- Effective traffic management improves overall road safety.

Contributing Factors



Road Safety Insight

Speed management, road infrastructure improvements, and better visibility represent the greatest opportunities to reduce severe road collisions.

When Did It Happen? – Temporal Analysis

Traffic Collision Trends

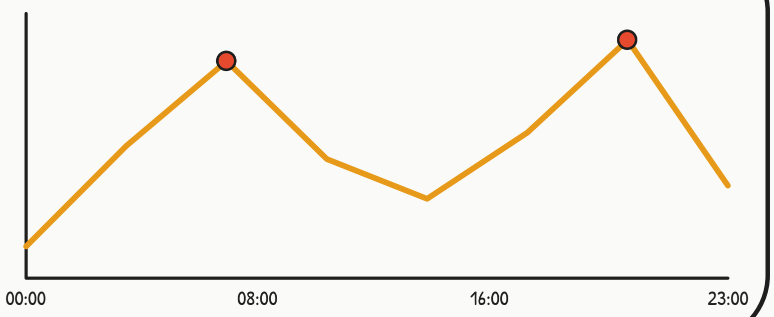
Road collisions follow consistent daily and seasonal patterns linked to traffic demand.

Key Findings

- Collision frequency increases during morning and evening commuting periods.
- Nighttime collisions are generally less frequent but more severe.
- Higher traffic volumes lead to increased collision occurrence during peak hours.
- Seasonal weather changes influence road safety throughout the year.

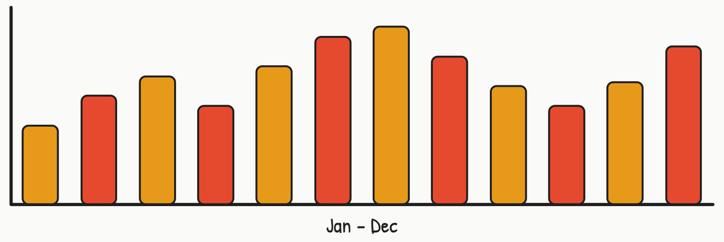
Road Safety Interpretation: Collision frequency is strongly related to traffic volume, while collision severity increases under reduced visibility and adverse environmental conditions. Preventive measures should therefore prioritize rush hours and nighttime periods.

Accidents by Hour



Suggested Visual: Line Chart – Accidents by Hour

Accidents by Month



Suggested Visual: Column Chart – Accidents by Month

Where Did It Happen? – Geographic Risk Analysis

Collision Hotspots: Geographic analysis identifies the locations where safety improvements can have the greatest impact.

High-Risk Areas

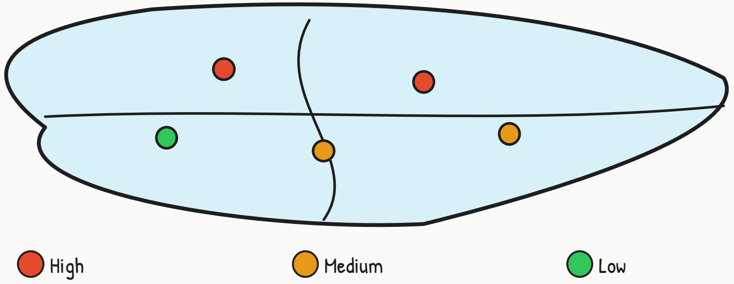
- Urban roads record the highest number of reported collisions.
- Major intersections and principal roads are recurring accident hotspots.
- Several local authorities consistently report higher casualty levels than surrounding areas.

Priority Areas

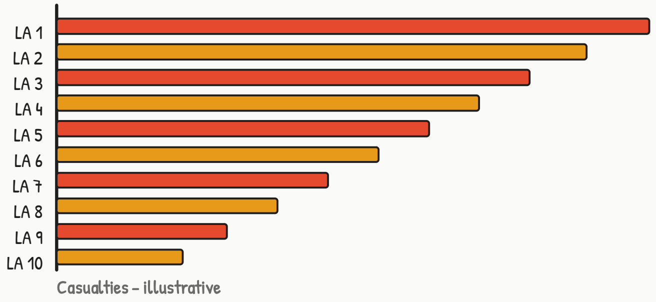
- Rural roads experience fewer collisions overall but tend to produce more severe crashes due to higher travel speeds.
- High-risk locations should receive priority for engineering improvements, speed enforcement, and traffic monitoring.

Road Safety Insight: Focusing resources on collision hotspots enables authorities to reduce accidents more efficiently and maximize the impact of infrastructure investments.

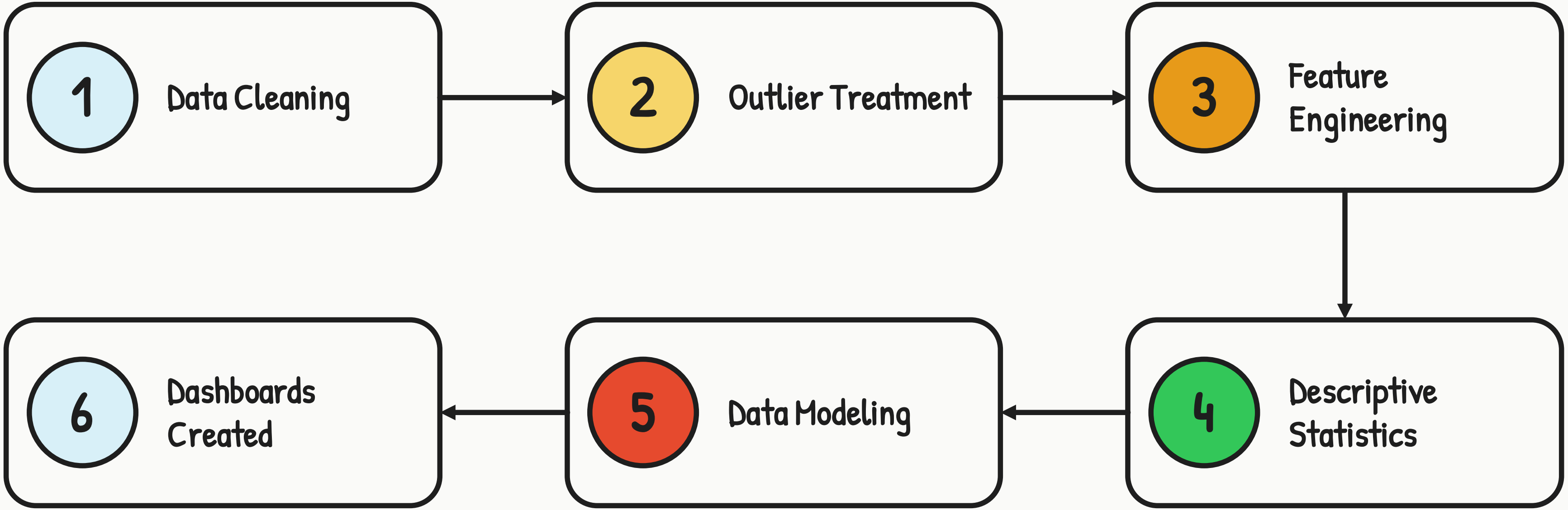
Filled Map – Collision Distribution by Region



Top 10 High-Risk Local Authorities



Project Summary



Project Summary

1 Data Cleaning

- Fixed data types (dates, numeric values, and categorical fields).
- Removed duplicate records and handled missing values.
- Standardized categorical values across all datasets.
- Validated data consistency between Collision, Casualty, and Vehicle tables.

2 Outlier Treatment

- Identified and reviewed unusual accident and casualty records.
- Validated extreme values before analysis.
- Preserved valid outliers representing real accident events.

Project Summary

3 Feature Engineering

- Created Year, Month, Quarter, and Day of Week from accident dates.
- Grouped speed limits into meaningful categories.
- Derived accident time periods (Morning, Afternoon, Evening, Night).
- Created severity categories and calculated accident-related KPIs.

4 Descriptive Statistics

- Analyzed accident, casualty, and vehicle distributions.
- Examined accident severity across different conditions.
- Summarized weather, road surface, and light condition frequencies.
- Identified trends and relationships among key variables.

Project Summary

5 Data Modeling

- Built a Star Schema with 3 Fact Tables and 13 Dimension Tables.
- Established one-to-many relationships using common reference keys.
- Organized dimensions for Date, Time, Location, Weather, Road Conditions, Vehicle, Driver, and Casualty attributes.
- Optimized the data model for efficient Power BI analysis and interactive reporting.

6 Dashboards Created

- **Excel:** 5 interactive dashboards with filters
- **Power BI:** 3 dashboards pages (Satisfaction, Drivers, Performance)
- **Tableau:** 4 views with drill-down capabilities
- **Python:** 4 visualization dashboards

Excel: Data Modeling & Descriptive Statistics

Descriptive Statistics Summary Table

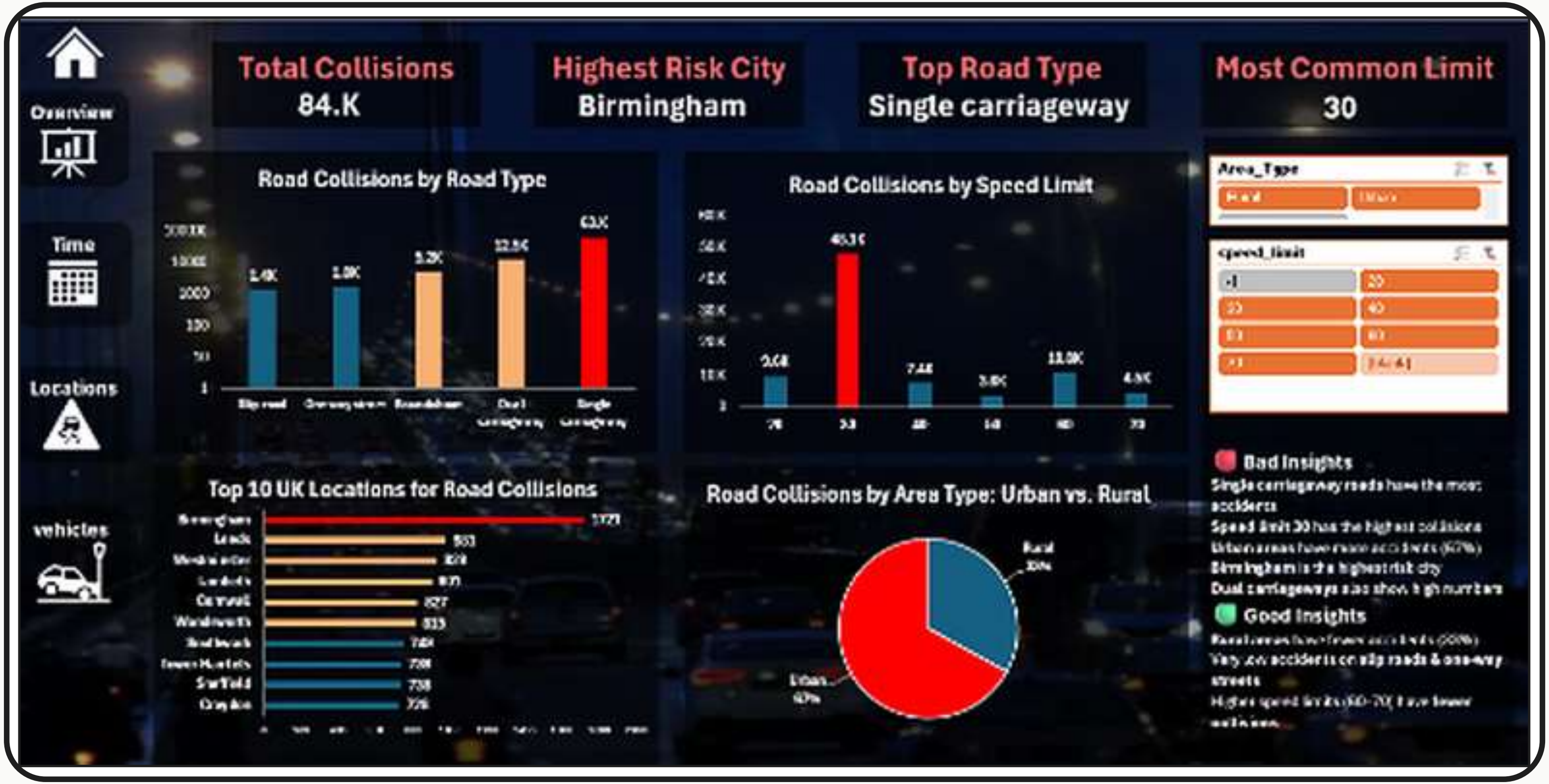
Metric	Max	Median	Min	Std Dev	Mean
Driver Age	101	34	-1	21.44	34.68
Casualty Age	101	34	-1	19.60	36.92
Speed Limit	70	30	-1	14.23	35.99
Number of Vehicles	26	2	1	0.69	1.83
Number of Casualties	70	1	1	0.70	1.27

Excel Dashboards

Cover → Dashboard 1: Executive Overview → Dashboard 2: Time Analysis → Dashboard 3: Road Analysis → Dashboard 4: People & Casualty Analysis

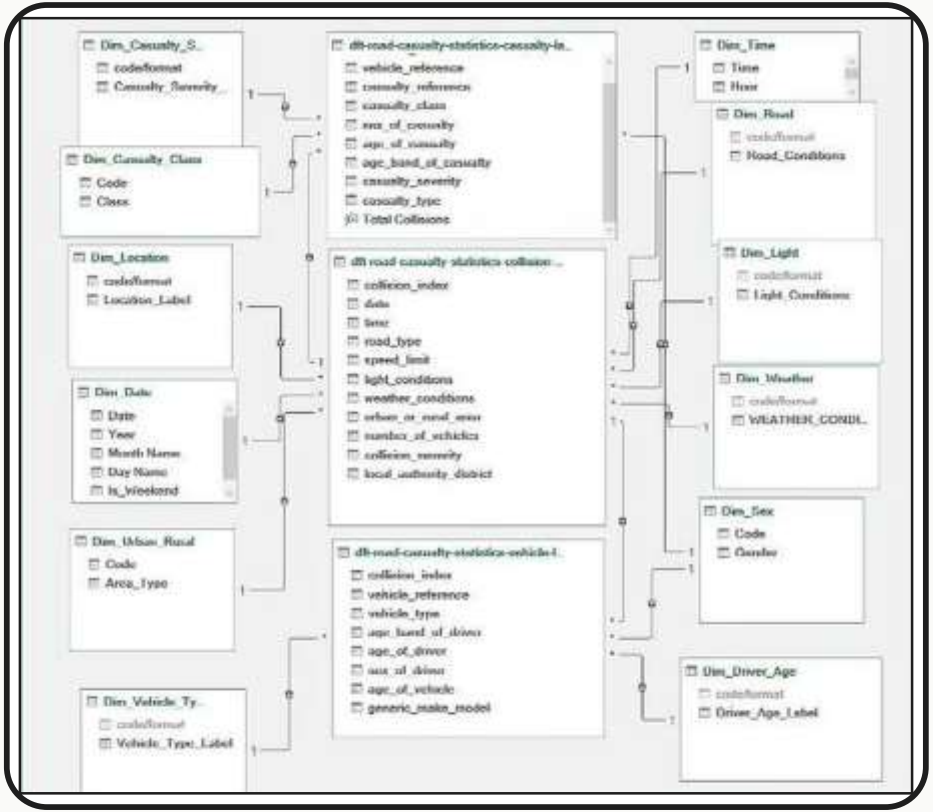
Excel Dashboards

Sample of the Excel analysis work — 1 / 4



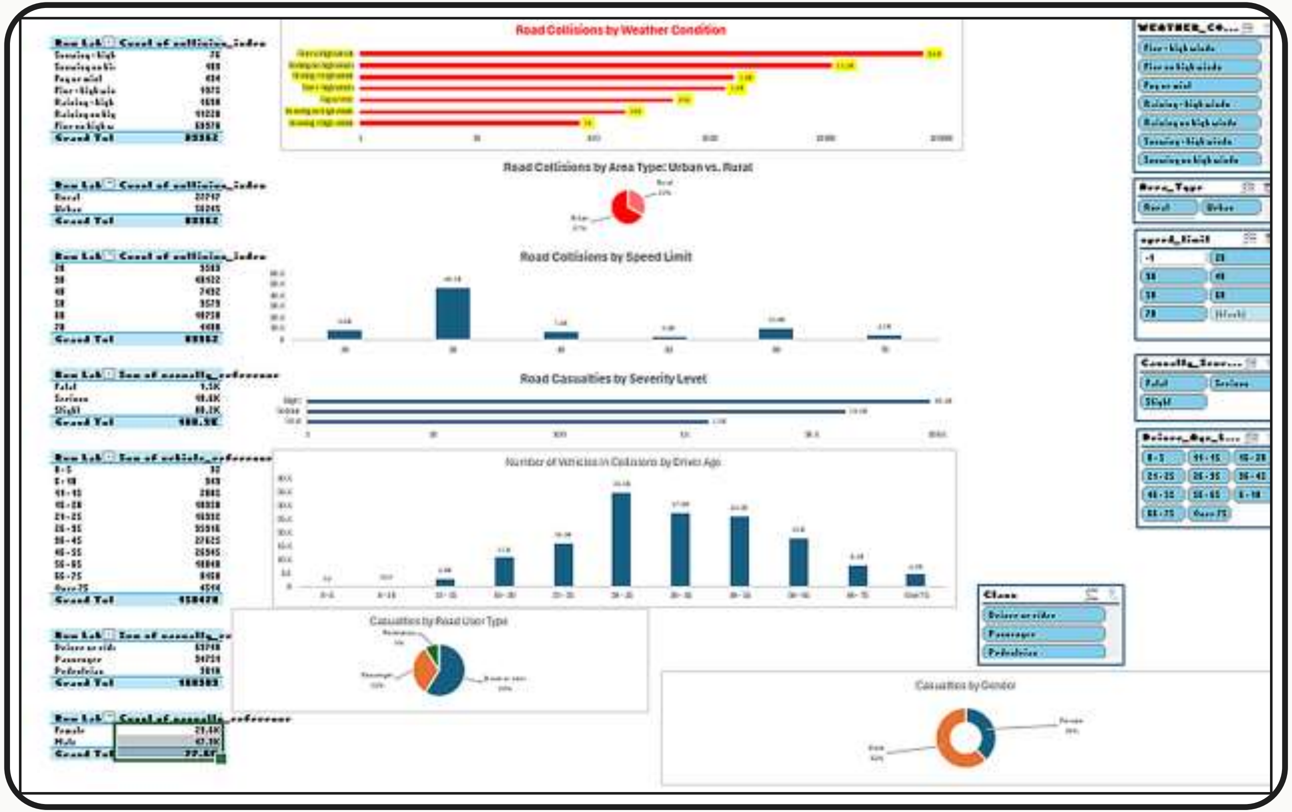
Excel Modeling

Sample of the Excel analysis work — 2 / 4



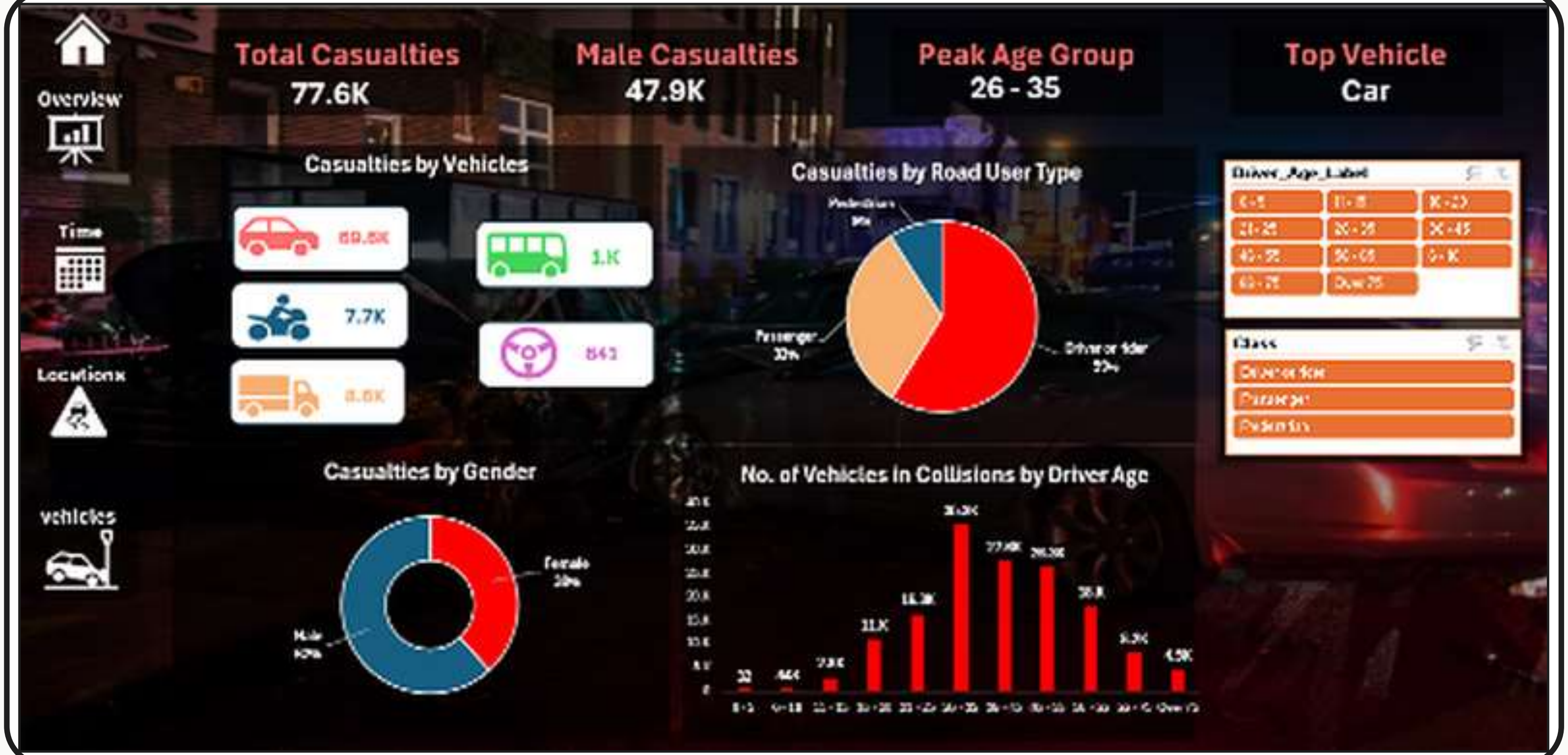
Excel Pivot Table

Sample of the Excel analysis work — 3 / 4



Excel Dashboards

Sample of the Excel analysis work — 4 / 4



SQL

Key Data Cleaning & Preprocessing Steps

01 Data Type Fixing & Standardization

- Converted accident date and time fields to appropriate date/time formats.
- Corrected numeric data types for age, speed limit, and reference fields.
- Standardized categorical values across weather, road, and light conditions.
- Renamed columns and applied consistent naming conventions.

02 Data Integration & Deduplication

- Joined Collision, Casualty, and Vehicle datasets using accident reference IDs.
- Removed duplicate records while preserving data integrity.
- Validated record counts before and after merging.
- Ensured referential integrity between related tables.

03 Missing Value Handling

- Identified missing and unknown values across all datasets.
- Replaced invalid codes with NULL where appropriate.
- Imputed selected missing values using suitable statistical methods.
- Verified data completeness before analysis.

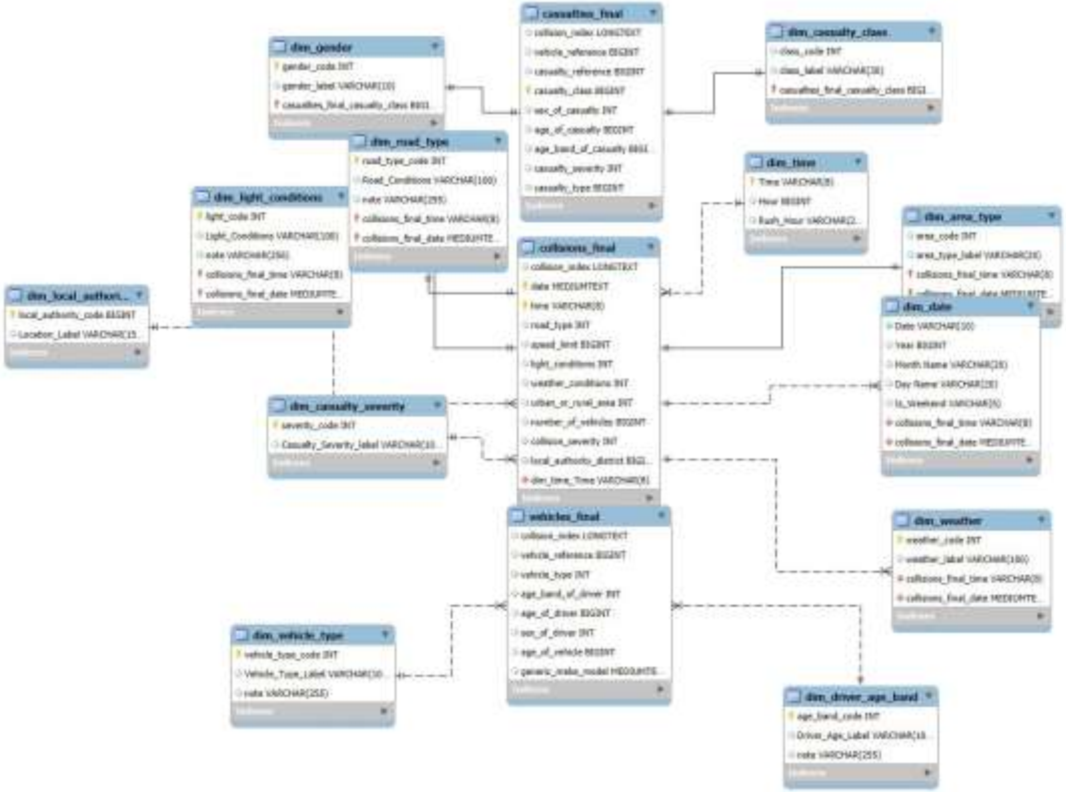
04 Outlier Treatment

- Reviewed unusual values in driver age, casualty age, and speed limit.
- Validated extreme records against accident data.
- Retained legitimate outliers representing real accident events.
- Documented data quality observations.

05 Feature Engineering

- Extracted Year, Month, Quarter, Day Name, and Hour from accident dates.
- Created Time Period categories (Morning, Afternoon, Evening, Night).
- Grouped driver and casualty ages into age bands.
- Derived analytical fields to support accident trend and severity analysis.

SQL: Star Schema & ERD



UK Road Safety Data Analysis

Project Overview & Methodology

OBJECTIVE

Analyze five years of UK collision records to uncover when, where, and under what conditions the most severe crashes occur — and turn those patterns into evidence-based road-safety recommendations.

DATASET

UK Government STATS19 open road-safety dataset — 503,475 collision records (2020–2024) covering severity, location, timing, weather, road and light conditions, and vehicle/casualty details.

TOOLS & LIBRARIES

- Pandas
- NumPy
- Matplotlib
- Seaborn
- SciPy

8 analytical stages — from raw data to boardroom-ready recommendations

ANALYSIS WORKFLOW

1. Data Collection
2. Data Cleaning
3. Preprocessing
4. Feature Engineering
5. Exploratory Analysis
6. Statistical Analysis
7. Dashboard Development
8. Business Recommendations

Python Analysis Process & Key Results

From data-quality checks to statistically grounded patterns

ANALYSES PERFORMED

- **Data Quality Checks**
  missing values · duplicates
- **Feature Engineering**
  time & severity features
- **Time-Based Trends**
  year · month · day · hour
- **Weather & Road Conditions**
  weather · surface · light
- **Speed & Location**
  speed limit · urban vs rural
- **Correlation Analysis**
  vehicles · casualties · speed

KEY NUMBERS

76.7%
Slight collisions

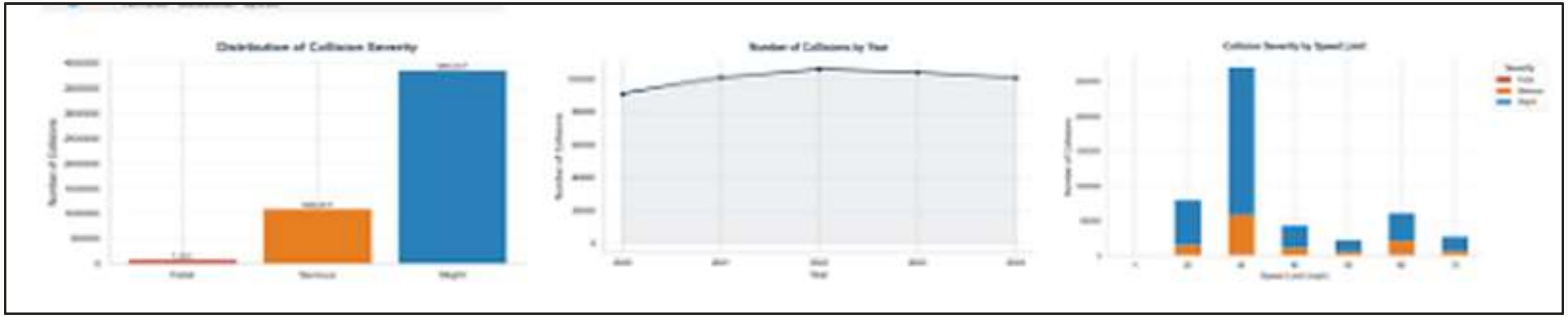
21.8%
Serious collisions

1.5%
Fatal collisions

2022
Peak collision year

17:00
Peak collision hour

60 mph
Highest fatal-share speed band (1.9%)



Statistical Analysis, Insights & Recommendations

Testing whether the visual patterns are statistically real ($\alpha = 0.05$)

χ^2 Chi-Square Test

H_0: Severity is independent of speed limit / weather / light / road surface.

Rejected $p < 0.005$ (all four factors)

Severity is significantly associated with each factor — though effect sizes are small (Cramér's V = 0.01–0.06).

F One-Way ANOVA

H_0: Mean casualties per collision are equal across Morning / Afternoon / Evening / Night.

Rejected $F = 107.98, p < 0.0001$

Casualty counts genuinely differ by time of day — highest at night (1.34 avg).

t Independent T-Test

H_0: Mean speed limit is equal for weekend vs. weekday collisions.

Rejected $t = 18.76, p < 0.0001$

Weekend collisions occur at significantly higher average speed limits (36.7 vs 35.8 mph).

KEY BUSINESS INSIGHTS

- Severity is overwhelmingly Slight (76.7%) — but Fatal collisions (1.5%) cause the most harm per incident (1.65 avg casualties).
- The 60 mph speed band is disproportionately deadly — 5.9% of its collisions are fatal, the highest of any speed limit.
- Collisions track exposure, not just danger: they peak in 2022, at 17:00, in Fine weather and Urban areas — simply where most driving happens.
- Numeric correlations are weak ($|r| \leq 0.21$); the real signal lives in categorical risk factors, confirmed by hypothesis testing.

RECOMMENDATIONS

1. Improve lighting on high-risk roads
2. Increase speed enforcement on high-speed roads
3. Focus safety campaigns on peak collision hours
4. Enhance road maintenance in adverse weather
5. Use predictive analytics for future accident hotspots

Data reflects collision counts rather than exposure-adjusted risk; results are based on the STATS19 open dataset (2020–2024).

Power BI: Dashboard 1 — Road Accident Overview

84K
Total Accidents

103K
Total Casualties

1K
Fatal Casualties

1.28%
Fatality Rate

Visualizations

- **Severity Breakdown** — Donut chart of Slight, Serious & Fatal casualties.
- **Accidents Over Time** — Line chart of the monthly accident trend.
- **Accidents by Road Type** — Bar chart comparing accidents across road types.

Filters & Slicers

- **Year** — filter the data by year.
- **Road Type** — filter by road category.
- **Weather Condition** — filter by weather conditions.

Power BI: Dashboard 1 — Road Accident Overview

Power BI: Dashboard 2 — Time & Environmental Conditions

84K	103K	1K	1.28%
Total Accidents	Total Casualties	Fatal Casualties	Fatality Rate

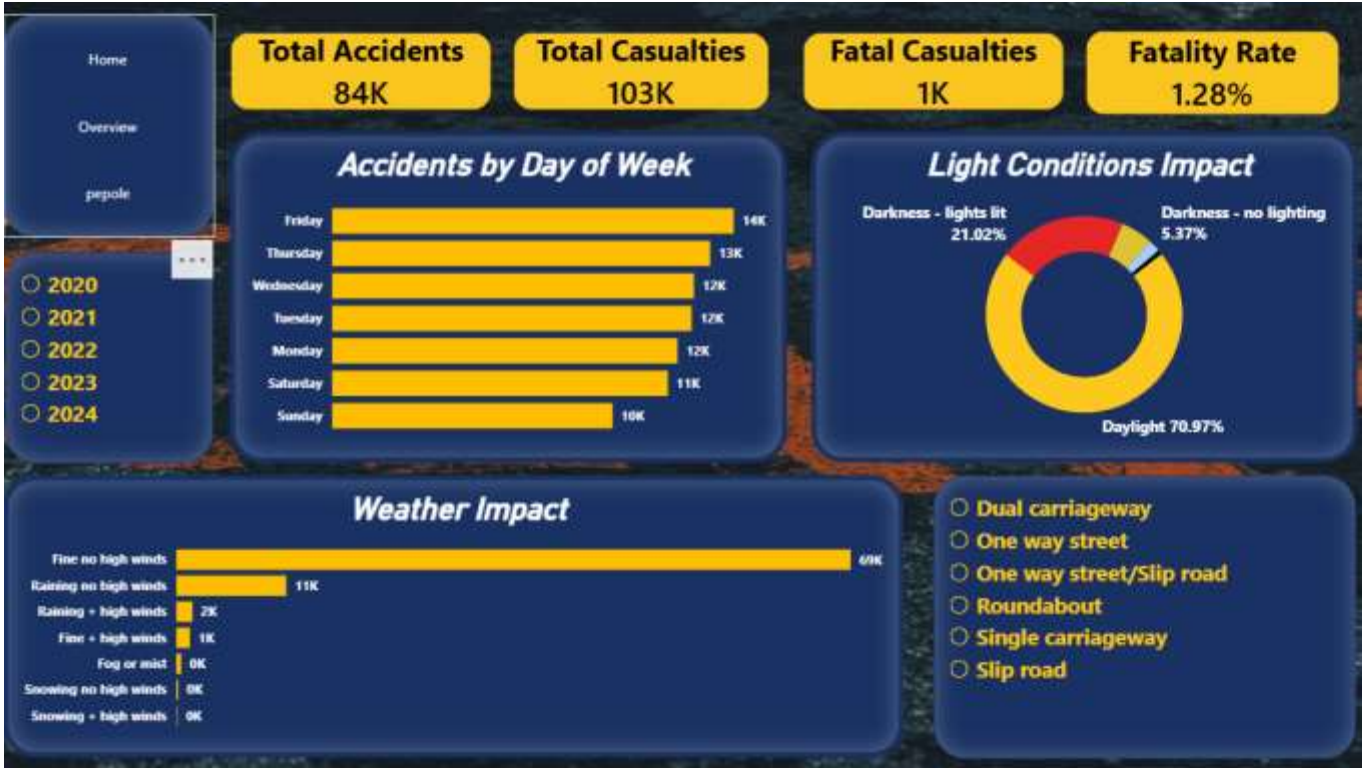
Visualizations

- **Accidents by Day of Week** — Bar chart comparing accident frequency across weekdays.
- **Light Conditions Impact** — Donut chart of accidents by daylight vs darkness.
- **Weather Impact** — Bar chart of accidents across weather conditions.

Filters & Slicers

- **Year** — filter the data by year.
- **Road Type** — filter accidents by road type.

Power BI: Dashboard 2 – Time & Environmental Conditions



Power BI: Dashboard 3 — Casualty & Demographic Analysis

103K
Total Casualties

Van / Goods 3.5T
Top Vehicle Type

1.28%
Fatality Rate

Visualizations

- **Age Group Distribution** — Treemap of casualties across age groups.
- **Casualties by Gender** — Decomposition tree breaking down casualties by gender.
- **Casualties by Road User** — Pie chart of Driver/Rider, Passenger & Pedestrian.
- **Light Conditions Table** — Casualties by different light conditions.

Filters & Slicers

- **Age Group** — filter the data by age category.
- **Gender** — filter the data by gender.

Power BI: Dashboard 3 – Casualty & Demographic Analysis

Tableau Data Modeling

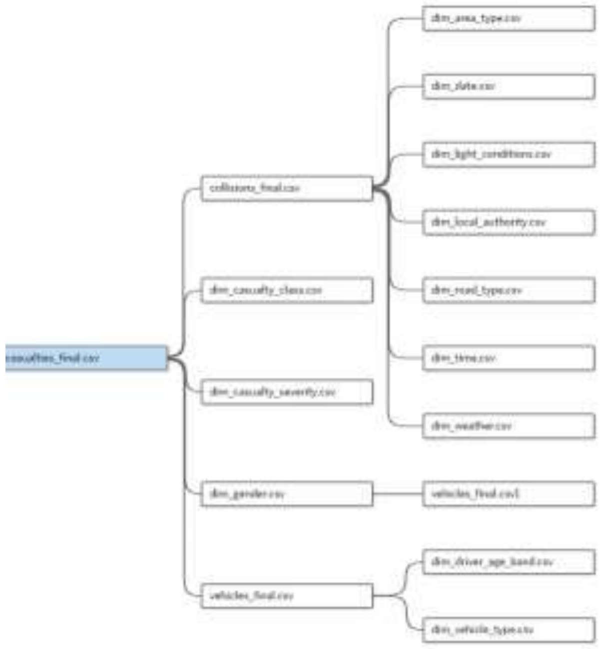


Tableau: Dashboard 1 – Road Safety Overview

Key metrics · visualizations · insights

83,971 Total Accidents	138,140 Total Casualties	3.5% Night Accident Rate	36.57 mph Average Speed Limit

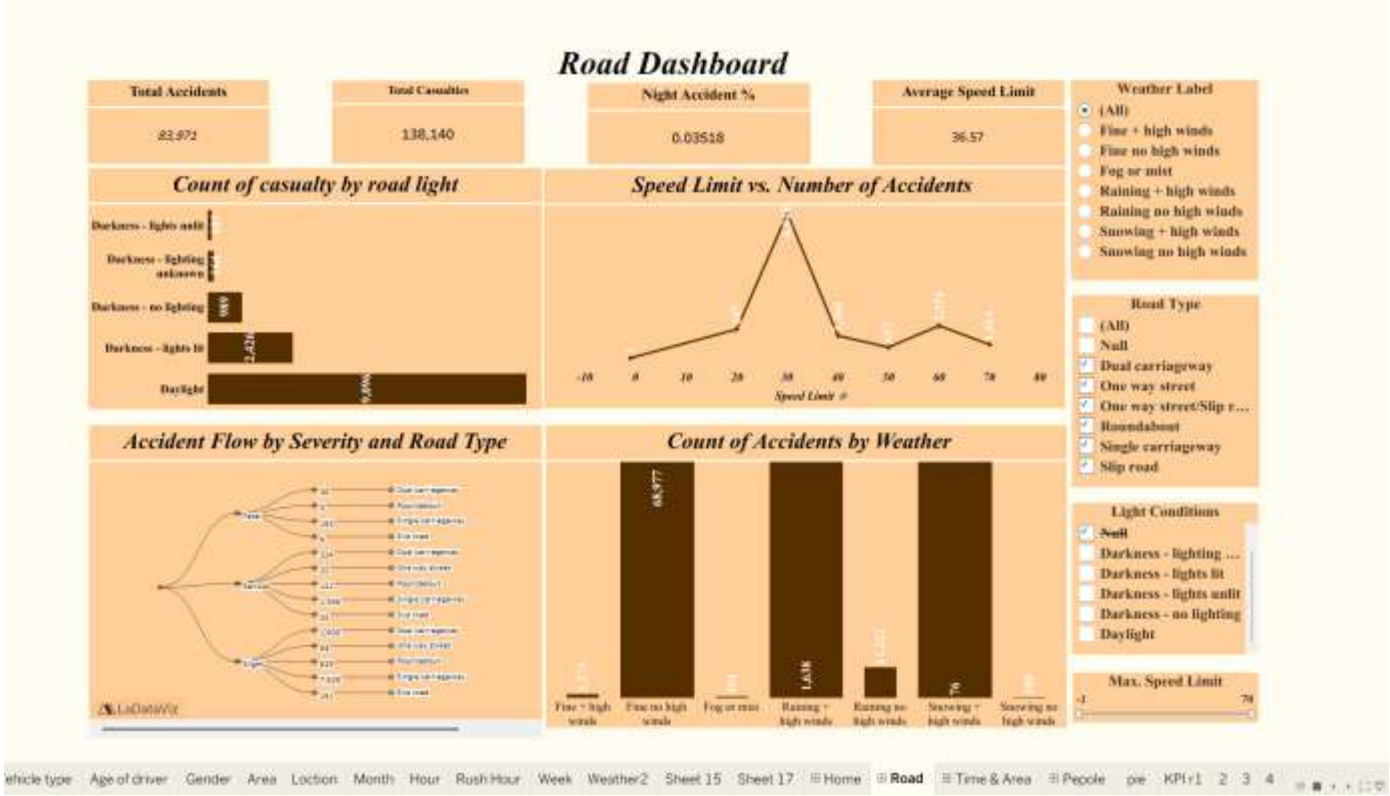
VISUALIZATIONS

- **Casualties by Road Light**
  Distribution of casualties under different lighting conditions.
- **Accidents by Speed Limit**
  Relationship between speed limits and accident frequency.
- **Severity by Road Type**
  Collision severity across different road types.
- **Accidents by Weather**
  Accident distribution under various weather conditions.

KEY INSIGHTS

- Most collisions occur in Daylight conditions.
- 30 mph roads record the highest number of accidents.
- Slight injuries account for the majority of casualties.
- Fine weather (No High Winds) has the highest accident count due to increased traffic volume.

Tableau: Dashboard 1 – Road Safety Overview



Thank You

Traffic Safety and Accidents — Data Analysis Project